

Microplastic Polymer Analysis Report

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Prediction Results

Detected Polymer	PP
Confidence Score	0.67%
Model Used	base_model

FTIR Spectral Analysis

- 1455 cm⁻¹ → CH bending
- 1375 cm⁻¹ → CH₃ symmetric deformation

Confidence Interpretation

■■■■■■■■■■ 0.67%

Scientific References

[1] Andrady, A. L. (2017). The plastic in microplastics: A review. Marine Pollution Bulletin, 119(1), 12–22. DOI: <https://doi.org/10.1016/j.marpolbul.2017.01.082>

[2] Fotopoulou, K. N., & Karapanagioti, H. K. (2012). Surface properties of beached plastic pellets. Marine Environmental Research, 81, 70–77. DOI: <https://doi.org/10.1016/j.marenvres.2012.08.001>

[3] Song, Y. K., Hong, S. H., Jang, M., Han, G. M., Jung, S. W., & Shim, W. J. (2015). Combined effects of UV exposure duration and mechanical abrasion on microplastic fragmentation by polymer type. Environmental Science & Technology, 49(7), 4368–4376. DOI: <https://doi.org/10.1021/es505863x>

[4] Pripke, S., Lorenz, C., Rascher-Friesenhausen, R., & Gerdt, G. (2017). An automated approach for microplastics analysis using FTIR microscopy and image analysis. Analytical Methods, 9(9), 1499–1511. DOI: <https://doi.org/10.1039/C6AY02476A>

[5] Jung, M. R., Horgen, F. D., Orski, S. V., Rodriguez, C. V., Beers, K. L., Balazs, G. H., Jones, T. T., Work, T. M., Brignac, K. C., Royer, S. J., Hyrenbach, K. D., Jensen, B. A., & Lynch, J. M. (2018). Validation of ATR FT-IR to identify polymers of plastic marine debris. Marine Pollution Bulletin, 127, 704–716. DOI: <https://doi.org/10.1016/j.marpolbul.2017.12.061>

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